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**Summary of problem**

GlobalVest Partners must decide whether to allocate 15% if its $1 billion portfolio into cryptocurrencies. This decision hinges on assessing the trade-off between potential returns and the inherent risks associated with this asset class. Market behavior under both options, whether to invest or not to invest, is uncertain, and GlobalVest Partners must weigh the expected monetary outcomes and model the outcomes using a specified utility function.

**Methods and Model**

In order to analyze the investment data, I constructed a decision tree with two primary paths: investing 15% of their portfolio in a diversified cryptocurrency or maintaining their current allocation. For each decision, there were different outcomes depending on the market performance probabilities.

Key information used:

- Investment portion: 15% of $1 billion = $150 million
- Risk tolerance (R): $50 million (base case)
- Scenarios if investing:
  - High Growth (30% chance): +30% return = $45M gain
  - Moderate Growth (50% chance): +10% return = $15M gain
  - Market Correction (20% chance): -20% return = -$30M loss
- Scenarios if not investing:
  - Benchmark Exceeds (20% chance): +20% return on existing portfolio = $30M gain
  - Matches Crypto (50% chance): +10% return = $15M gain
  - Underperforms (30% chance): 0% return = $0 gain

Key equations:

- Utility function: U(W)=1-e^(-W/R)
- Expected Value (EV): $EV = \sum p_i * R_i$
- Expected Utility (EU): $EU = \sum p_i * U(W_i)$
  - W is the investment gain, R is the risk tolerance, and p is the probability.

Excel was used to simulate this model. Input values were embedded directly into the formulas used within a base case analysis as well as within a Goal Seek analysis.




Figure 4: Excel spreadsheet of decision tree

**Questions**

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1. Construct a decision tree illustrating GlobalVest Partners' decision and the possible outcomes. Clearly label all nodes, branches, probabilities, and potential gains at the end of the year for each path.
    a. See Figure 4 above. The decision tree includes two decision nodes branching into respective chance nodes with assigned probabilities and resulting monetary gains. Utility and expected utility are calculated at each terminal node. Excel snapshot tables summarize each path and associated values.
2. Calculate the Expected Value (EV) of the portfolio gains at the end of the year for both strategies (Invest in Cryptocurrency and Do Not Invest). Based solely on expected value, which decision should GlobalVest Partners make?
    a. Based on the decision tree analysis, if they invest, they should have an expected value gain of $15 million, but if they do not invest, then their expected value gain is only $13.5 million. Based on the outcome of the expected value alone, they should invest.
3. Calculate the Expected Utility (EU) for both strategies using the provided utility function.
    a. Based on the decision tree analysis, if they invest, their expected utility value is 0.1432, but if they do not invest, then their utility value is 0.2198. Based on the utility value alone, they should not invest.
4. At what risk tolerance value, are both strategies equivalent?
    a. Using the Goal Seek function in Excel, I adjusted the risk value until the expected utility values equaled. The calculated risk tolerance turned into $186,253,318.
5. Based on the expected utility, and considering GlobalVest Partners' risk aversion as represented by the utility function, which decision should they make? Explain why the decision based on expected utility might differ from the decision based on expected value.
    a. Expected utility captures the company's risk aversion more appropriately than expected value. Given what the data showcases, they should not invest in cryptocurrency under the given risk preference of $50 million.

**Executive Summary**

GlobalVest Partners must choose between maintaining their current portfolio allocation or adjusting 15% into cryptocurrency. While cryptocurrency offers a slightly higher monetary return, the utility value analysis reveals that it poses a much higher risk than their current risk tolerance. Based on the expected utility, GlobalVest Partners should not move forward with investing into cryptocurrency.